

TOTAG GROUP OF COMPANIES LTD

TECHNICAL PROPOSAL

Development of a Digital Farmer Registry System in Liberia

A secure, interoperable and nationally owned platform connecting agriculture, social protection and service delivery

Solicitation	Bidder
FAO RFP No. 2026/FRLIR/FRLIR/137641	TOTAG GROUP OF COMPANIES LTD
Beneficiary country: Liberia	UNGM ID: 1185811
Submission deadline: 4 August 2026, 12:00 Monrovia time	HP Partner ID: 991002750397
Proposal type: Technical — no financial information	Contact: info@totaggroup.com

Prepared for
Food and Agriculture Organization of the United Nations — Liberia

CONFIDENTIAL TECHNICAL OFFER This document contains no price or other financial proposal information. Prescribed FAO forms and documentary evidence are to be included in the final Technical Proposal file without alteration to their required declarations.

Submission draft • 26 July 2026

Letter of Technical Proposal

26 July 2026

To: Procurement Unit, FAO Representation in Liberia

Subject: Technical Proposal — Development of a Digital Farmer Registry System in Liberia, RFP No. 2026/FRLIR/FRLIR/137641

Dear Evaluation Committee,

TOTAG GROUP OF COMPANIES LTD respectfully submits this Technical Proposal to design, develop, test, deploy and transfer a secure, interoperable Digital Farmer Registry (DFR) and integrated Management Information System for Liberia. Our approach is built around national ownership, offline-first field operations, authoritative geospatial farmer and parcel records, privacy-by-design, standards-based interoperability, and practical capacity transfer to the Ministry of Agriculture and partner institutions.

We understand that the assignment is not simply a software build. It is an institutional modernization programme requiring assessment of existing systems, data and governance design, operational SOPs, interoperable technology, capacity development, pilot validation, national rollout and documented learning. We therefore propose a seven-month, stage-gated delivery approach aligned to the maximum period stated in the Terms of Reference, subject to FAO's written resolution of the inconsistent five-, six- and seven-month references elsewhere in the RFP.

TOTAG brings Liberian leadership and delivery presence, digital-government and workflow expertise, enterprise software engineering capability, donor-linked operations and training experience, and a registered HP Partner Portal relationship. Our implementation model combines senior technical architecture with county-level stakeholder engagement and a disciplined evidence, testing and knowledge-transfer process.

We confirm that this Technical Proposal contains no financial information and will remain valid for three months from the tender opening date. We accept that any contract award remains subject to FAO's rules, due diligence, clarification and the final contract.

Yours faithfully,


Alpha Gwoah


Mrs. Tonieh Alpha Gwoah
Deputy Chief Executive Officer (D-CEO)
Authorized Signatory
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Document Control

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1. Executive Summary

Liberia's agricultural and social-protection institutions need a trusted, shared view of farmers, households, farms and parcels that can operate reliably in low-connectivity environments and exchange data safely with national systems. TOTAG proposes an open, modular and government-owned DFR that converts this need into an operational national service—not a stand-alone database.

OUR DELIVERY PROMISE One verified farmer record; one traceable parcel history; multiple authorized services. The platform will support targeting, extension, inputs, vouchers, payments, monitoring and policy analysis while enforcing consent, least-privilege access, auditability and national data sovereignty.

- Institution first: governance, SOPs, roles, data stewardship and change adoption are designed alongside the technology.
- Offline first: field registration, GPS parcel capture, validation and synchronization are resilient to intermittent connectivity.
- Interoperable by design: versioned REST APIs, OpenAPI documentation and standards-based exchange support social protection, identity, statistics, land, weather, payments and partner systems.
- Security by design: encryption, identity and access management, audit trails, separation of duties, backup, disaster recovery and incident response are designed into every layer.
- Evidence-led rollout: discovery, prototype validation, pilot, controlled migration and release gates reduce implementation risk before nationwide deployment.
- Sustainable ownership: administrators, developers, analysts, enumerators and help-desk teams receive role-specific training, manuals, source code, configurations and operational runbooks.

1.1 Strategic Outcomes

Outcome	How the proposal delivers it	Evidence of success
Trusted national registry	Canonical farmer, household, farm and parcel model; deduplication; validation and approval workflows	Duplicate rate, completeness, validation turnaround, audit exceptions
Better service delivery	Eligibility, programme enrolment, vouchers, inputs and payment exchange interfaces	Traceable beneficiary lifecycle and reconciliation
Field reach	Android-compatible progressive/mobile application with offline storage, GPS and queued synchronization; USSD/SMS options	Sync success, registration throughput, device coverage
Government insight	Role-based dashboards, GIS layers, production and vulnerability analytics	Timely county/district reports and exportable indicators
National ownership	Government-hosted/hybrid deployment, source/configuration handover, training and SOPs	Trained users, accepted runbooks, successful recovery exercise

2. Understanding of the Assignment

The Terms of Reference require seven connected components: assessment of existing registries and social-protection systems; DFR and governance design; SOP development; software development/customization and interoperability; capacity building; testing and national rollout; and documentation of lessons and best practices. Each component is necessary to deliver a platform that is technically functional, institutionally legitimate and operationally sustainable.

2.1 Liberia-Specific Delivery Context

- Farmer and farm information is distributed across programmes and institutions, creating duplication, inconsistent definitions and limited reuse.
- County and district users need tools that remain productive when connectivity is absent or unreliable.
- Parcel geolocation and boundary capture must work with field GPS accuracy, retain provenance and support later quality correction.
- Data exchange with social-protection and government systems requires governance agreements, data-minimization rules and secure interfaces—not only API code.
- Women, youth, persons with disabilities and vulnerable households must be visible in disaggregated reporting without exposing sensitive personal data.
- National rollout requires institutional ownership, a help-desk model, field supervision, training-of-trainers and measurable adoption.

2.2 Compliance with the Required Scope

RFP component	TOTAG response	Primary output
1. Assess existing systems	Stakeholder mapping, inventory, interviews, data/process review, infrastructure and readiness assessment	Gap Assessment Report
2. Design DFR	Business, data, application, integration, infrastructure, security and governance architecture	System Design Document
3. Develop SOPs	Role-based operating procedures, RACI, controls, forms, service levels and escalation paths	SOP Manual
4. Develop/customize	Modular web/mobile platform, GIS, analytics, integration layer, database and administration	Functional interoperable platform
5. Build capacity	Training needs analysis, training-of-trainers, role-based practical sessions and coaching	Materials and Training Reports
6. Test and roll out	Quality assurance, security tests, pilot, defect resolution, migration and controlled national deployment	Pilot and Rollout Report
7. Capture learning	Performance review, user feedback, lessons, sustainability and policy recommendations	Final Report

3. Technical Approach and Methodology — TECH-3

3.1 Delivery Principles

- Co-design with the Ministry of Agriculture, MGCSP, CDA, LISGIS, county/district teams, programmes and farmer representatives.
- Use open standards and avoid unnecessary vendor lock-in while meeting the specified enterprise-server requirement.
- Apply privacy, security, accessibility and inclusion from requirements through operations.
- Configure before customizing; customize only where Liberia's approved process requires it.
- Validate each stage through documented acceptance criteria and traceable requirements.
- Transfer knowledge continuously rather than deferring handover to the end.

3.2 Phase 1 — Mobilization, Discovery and Gap Assessment

Within the opening weeks, TOTAG will establish governance, confirm the system boundary, reconcile RFP ambiguities and build a verified baseline. Discovery will combine document review, stakeholder interviews, system demonstrations, field-process observation, data profiling and infrastructure assessment.

- Inception workshop, stakeholder map, governance calendar, RAID log and approved communication plan.
- Catalogue of farmer registries, social-protection MIS, datasets, interfaces, data owners, legal bases, identifiers and data-quality conditions.
- Field-registration journey mapping from community mobilization through verification, approval, service delivery, grievance and update.
- Infrastructure, connectivity, device, hosting, cybersecurity, backup and support-readiness assessment.
- Gap prioritization by operational impact, risk, feasibility, dependency and cost category, without including prices in this Technical Proposal.

3.3 Phase 2 — Requirements, Architecture and Governance Design

TOTAG will convert the baseline into an approved, testable design. A requirements traceability matrix will map each TOR requirement to user stories, data entities, interface specifications, controls, tests and deliverables. Architecture decisions will be recorded with alternatives and rationale.

Architecture layer	Proposed specification
Experience	Responsive web portal; offline-capable mobile field application; optional USSD/SMS services; WCAG-informed accessibility and low-bandwidth design
Application	Modular services for registry, household, farm/parcel, crops/livestock/equipment, programmes, verification, grievances, extension, vouchers/inputs/payments, analytics and administration
Geospatial	WGS84 GPS points/polygons, guided parcel walk, satellite/basemap visualization, GeoJSON, spatial validation, area calculation, overlap/topology checks and controlled correction history
Data	Relational transactional store with spatial extension, master/reference data, immutable audit events, document/object storage, analytics views and metadata catalogue
Integration	API gateway; REST/JSON and XML where required; OpenAPI 3.x; OAuth 2.0/OIDC or approved government identity; webhooks/queues; throttling and versioning
Security	TLS 1.2+ in transit, AES-256-equivalent encryption at rest, MFA for privileged users, RBAC/attribute scopes, secrets management, centralized logging and secure SDLC
Infrastructure	HP Gen10/11 or equivalent enterprise platform sized after assessment; virtualization/containers; high availability, monitoring, encrypted backup, replication and disaster recovery
Operations	Automated deployment, environment separation, configuration management, monitoring, alerting, runbooks, support workflows and auditable release management

3.4 Farmer, Farm and Parcel Data Model

The canonical data model will separate a person, household, farm enterprise and parcel so that identities and relationships can evolve without corrupting history. Each record will carry provenance, consent/legal basis, collection device/user, timestamps, validation status and change history.

- Farmer: demographic and contact data, national or programme identifiers, gender, age/youth status, disability information where lawfully required, vulnerability and consent.
- Household: representative, members, relationships, dependency and social-protection references.
- Farm: ownership/tenure, location, facilities, equipment, labour and production profile.

- Parcel: GPS point or polygon, computed area, crop/season/use, tenure evidence, accuracy metadata, overlap and geometry-quality flags.
- Production: crops, livestock, equipment, inputs, harvest and market information with season and source.
- Institutional entities: cooperative, agribusiness, extension structure, programme, service provider and administrative geography.
- Service events: assessment, eligibility, enrolment, input/voucher/payment/extension transaction, grievance, verification and outcome.

3.5 Identity Resolution and Duplicate Management

Duplicate prevention will use a tiered process: deterministic checks on trusted identifiers and contacts; configurable probabilistic matching on names, dates, household and geography; reviewer work queues; merge/link rules; and a reversible, fully audited resolution process. Biometrics will not be introduced unless specifically authorized through legal, ethical, security and operational assessment.

3.6 GIS and Parcel Mapping

Enumerators will be able to pinpoint a farm, capture a GPS point, or walk/digitize a parcel boundary against an approved map provider or government basemap. The application will cache required map areas where licensing permits, show GPS accuracy before capture, prevent accidental self-intersections, calculate area, flag improbable or overlapping geometry, and retain both the original and approved geometry. Geocoding and reverse geocoding will support locality entry; boundaries will be validated against official county and district layers.

MAP PROVIDER DECISION Google Maps may be configured where licensing, offline availability, data-protection and government procurement requirements permit. The solution will isolate map-provider dependencies so the Government can use approved alternatives such as government geoservices or open-source basemaps without redesigning the registry.

3.7 Offline Field Registration and Synchronization

- Encrypted local storage, device binding, session timeout and remote revocation for managed devices.
- Configurable forms, reference-data packages and validation rules available offline.
- Queued create/update operations with idempotency keys, retry logic and visible synchronization status.
- Conflict handling based on field ownership, record version and supervisor review—never silent overwrite.
- Attachment compression, resumable transfer, sync telemetry and offline audit capture.
- Enumerator assignment, workload, route and supervision functions with privacy-preserving location controls.

3.8 Interoperability

The integration layer will exchange only the minimum approved data and will enforce purpose, authorization, schema validation, encryption, throttling and audit logging. Each interface will have an owner, data-sharing basis, field mapping, frequency, error process, service level and version lifecycle.

Target domain	Typical exchange	Control
National identity/civil registration	Identity verification and identifiers	Consent/legal basis, match confidence, minimal response
National Social Registry and cash-transfer MIS	Household vulnerability, programme enrolment and status	Data-sharing agreement, purpose limitation, reconciliation
Land, census, statistics, weather	Administrative boundaries, parcels, indicators and advisories	Authoritative-source metadata and versioning
Banks, MNOs and payment systems	Validated beneficiary/payment instruction/status	Tokenization, segregation of duties, reconciliation; no storage of unnecessary financial credentials

Target domain	Typical exchange	Control
Partner systems	Programme eligibility, service delivery and aggregate reporting	Tenant/programme scope, least privilege and export controls

3.9 Security, Privacy and Data Protection

A security and privacy workstream will maintain the asset inventory, data classification, threat model, risk register, control matrix and incident plan. Controls will be aligned with Liberia's applicable requirements, FAO contractual requirements and recognized security practices, with final compliance mappings confirmed during inception.

- Least privilege with configurable organization, programme, county, district, function and data-sensitivity scopes.
- MFA and stronger session controls for administrators, payment/voucher roles, auditors and remote privileged access.
- Separation of duties for registration, verification, approval, payment, reconciliation and administration.
- Encryption in transit and at rest; managed keys/secrets; secure backups; hardened operating systems and databases.
- Immutable audit trail for view, create, update, export, approval, merge, payment and administrative events.
- Secure coding, dependency scanning, code review, SAST/DAST, penetration testing and remediation before production.
- Data retention, archival, subject-request, correction, consent/notice, breach handling and secure disposal procedures.
- Restricted exports, masking of sensitive fields, watermarking where appropriate and anomaly alerts for bulk access.

3.10 Role-Based Access and Dashboards

Permissions will be policy-driven rather than hard-coded. A user receives one or more roles, each constrained by organization, programme, geography, function and data classification. Temporary elevation requires approval and expiry. The following dashboard families will be refined through user research:

User family	Primary dashboard tools
Farmer/household/cooperative	Profile, farms/parcels, programme/service status, consent, notices, grievances and controlled updates
Enumerator/senior enumerator	Assignments, offline forms, map capture, sync queue, quality checks, team workload and correction tasks
County/district/extension	Geographic caseload, verification, visits, production, advisories, service referrals and coverage maps
Programme/voucher/input/payment	Eligibility, approval queues, issuance/distribution/payment batches, exception handling and reconciliation
Verification/M&E/GIS/analyst	Quality queues, indicators, maps, datasets, metadata, analysis and privacy-controlled exports
Help desk/admin/security/audit	Tickets, user/role administration, configuration, health, logs, incidents, access reviews and read-only evidence
Development partner/oversight	Programme-scoped or aggregated dashboards and reports; identifiable data only where explicitly authorized

3.11 SOP Development

SOP family	Minimum content
Registration and update	Mobilization, consent/notice, identity, household/farm/parcel capture, evidence, correction and re-registration
Validation and approval	Automated checks, supervisor review, duplicate resolution, approval authority, rejection and appeal
Data management	Definitions, stewardship, quality rules, access, retention, backup, restoration, archival and disposal

SOP family	Minimum content
Integration	Onboarding, data-sharing authorization, interface testing, credentials, monitoring, change control and decommissioning
Service delivery	Eligibility, voucher/input/payment controls, distribution evidence, reconciliation and exception management
Grievance and support	Intake, classification, safeguarding, escalation, response targets, closure and feedback
Security and incidents	Account lifecycle, access review, vulnerability handling, incident triage, breach response and evidence preservation
Monitoring and reporting	Indicator definitions, reporting calendar, data-quality review, dissemination and confidentiality

3.12 Quality Assurance, Testing and Acceptance

Acceptance will be evidence based. Tests will trace directly to approved requirements, with severity thresholds and documented closure. Production deployment will not proceed while critical security or data-integrity defects remain open.

- Unit, API, integration, database migration, workflow and regression testing.
- Offline, sync conflict, low-bandwidth, device and browser compatibility testing.
- GIS accuracy, geometry, area and administrative-boundary validation.
- Performance, concurrency, load, endurance, backup restoration and disaster-recovery testing.
- Security testing, access-control verification, vulnerability scanning and independent penetration testing where contracted.
- User acceptance testing by representative national, county, district, field and partner users.
- Accessibility, usability, language/content and training-readiness checks.

3.13 Pilot and National Rollout

Pilot locations and sample sizes will be agreed during inception using criteria that represent connectivity, agricultural systems, geography, programme participation and institutional readiness. The pilot will validate end-to-end registration, verification, synchronization, integrations, service workflows, support and reporting. A formal go/no-go review will approve remediation and phased national deployment.

7. Prepare infrastructure, reference data, devices, user accounts, training and support.
8. Migrate approved pilot data through profiled, reconciled and reversible scripts.
9. Run supervised field operations and daily issue triage.
10. Measure adoption, quality, performance, security and user feedback.
11. Close priority defects and approve the rollout readiness report.
12. Deploy by agreed waves, with hypercare and county/district support.
13. Transition operations to government teams under an agreed responsibility matrix.

4. Work Plan and Deliverables – TECH-7

SCHEDULING BASIS The detailed plan uses seven months because the Terms of Reference and qualification section state a maximum of seven months. The RFP also contains inconsistent five- and six-month references; TOTAG will adopt FAO's written clarification without changing the scope.

Month	Primary activities	Milestone / deliverable
M1	Mobilization; inception; stakeholder and system inventory; discovery; field and infrastructure assessment	D1 Inception Report and approved work plan

Month	Primary activities	Milestone / deliverable
M2	Gap analysis; requirements; data governance; target operating model; architecture and prototype	D2 Gap Assessment; design review
M3	System design completion; SOP drafting; platform configuration/development; API and data model	D3 System Design Document; SOP draft
M4	Module development; offline/GIS; integrations; dashboards; security hardening; training design	D4 SOP Manual; alpha platform
M5	System/integration/security/performance testing; data preparation; administrator and trainer enablement	D5 Functional interoperable platform; training materials
M6	Pilot deployment; UAT; field coaching; measurement; remediation; rollout readiness	D6 Pilot Report; acceptance actions
M7	Phased national rollout; hypercare; knowledge transfer; lessons and sustainability planning	D7 Training reports; Final Report and handover

4.1 Deliverable Acceptance Package

Deliverable	Acceptance evidence
Inception Report	Approved methodology, governance, stakeholder plan, detailed schedule, assumptions and requirement-control process
Gap Assessment	Verified system/data inventory, process maps, infrastructure/security findings and prioritized recommendations
System Design Document	Approved requirements, architectures, data dictionary, interface catalogue, security/privacy design and sizing assumptions
SOP Manual	Version-controlled, role-specific procedures with RACI, controls, forms, service levels and approval
Functional Platform	Source/configuration, deployment package, test results, API documentation, admin guides, licenses/dependencies and defect status
Training Package	Needs analysis, curricula, manuals, exercises, attendance, assessments, feedback and coaching records
Pilot Report	Pilot scope, metrics, defects, remediation, user feedback, acceptance decision and rollout readiness
Final Report	Rollout status, results, lessons, sustainability, policy recommendations, asset register and signed handover

5. Project Governance and Management

A joint governance structure will maintain rapid decisions, clear accountability and traceable acceptance. TOTAG proposes a Steering Committee for strategic decisions, a Product and Technical Working Group for requirements and acceptance, and an Implementation Team for daily delivery.

Forum	Participants	Cadence	Purpose
Steering Committee	FAO, MoA and designated institutions; TOTAG Project Director/Manager	Monthly / stage gates	Scope, policy, dependencies, risks and acceptance

Forum	Participants	Cadence	Purpose
Product & Technical Group	Product owners, data/GIS/security/integration focal points, TOTAG leads	Weekly	Requirements, design, demos, data and integration decisions
Implementation Team	TOTAG workstream leads and government counterparts	Daily/biweekly	Delivery coordination, blockers, tests and rollout
Security & Data Governance	Data controllers/stewards, security, legal/privacy and audit focal points	Biweekly / as needed	Data use, access, risk, incidents and interfaces

5.1 Reporting and Control

- Weekly progress update: completed work, next actions, decisions, risks, issues and dependencies.
- Monthly progress report: deliverables, schedule, quality, staffing, change requests and outcome indicators.
- Requirements traceability, decision, change, risk/issue, defect and acceptance registers.
- Demonstrations at least every two weeks during development.
- Formal gate reviews for inception, design, alpha, pilot readiness, rollout readiness and handover.

5.2 Risk Management

Risk	Response	Owner / trigger
Delayed access to systems/data	Confirm focal points and access plan in inception; use interface mocks; escalate against decision dates	Joint; missed access milestone
Poor or duplicate legacy data	Profile, cleanse, map, reconcile, preserve provenance and require owner approval before migration	Data Lead; quality below threshold
Low connectivity/device limitations	Offline-first workflow, compression, resumable sync, device matrix and representative field testing	Technical Lead; sync failure trend
Unresolved legal/data-sharing basis	Data-flow inventory, minimum-data design, draft agreements and governance gate before live exchange	Government/FAO; interface blocked
Cybersecurity incident	Hardened environments, monitoring, response runbook, escalation, containment and recovery exercises	Security Lead; alert/incident
User adoption resistance	Co-design, champions, role-based training, job aids, coaching and adoption metrics	Change Lead; usage below target
RFP duration ambiguity	Plan to seven-month TOR maximum; rebaseline promptly after written FAO clarification	Project Manager; clarification issued
Key-expert evidence gap	Submit strengthened FAO-format CVs and documentary evidence or nominate compliant replacements before deadline	Bid Director; compliance review

6. Team Composition and Assignments — TECH-4

The proposed core team aligns each available expert to the role in which the documented experience provides the strongest contribution. Varnie N'Jola Karmo brings public-sector software project and technical leadership for the Indiana General Assembly; M. Tarwoyoubekowu Gwoah brings custom software, workflow and digital-government implementation capability; and Tonieh Alpha Gwoah brings beneficiary training, programme-data oversight, user support and Liberian field operations.

Proposed person / position	Principal responsibility	Evidence-based strength
Varnie N'Jola Karmo — Key Expert 1: Project Manager / Software Project Technical Lead	Project planning, technical leadership, requirements, architecture oversight, migration, quality, risk, delivery and acceptance	BSc Computer Information Systems; 25+ years enterprise software; since 2007 supporting mission-critical Indiana General Assembly systems; React, PostgreSQL, AWS, Oracle, Neo4j and GIS integration
M. Tarwoyouberkowu Gwoah — Key Expert 2: Software Developer / Digital Systems Implementation Lead	Custom application development, workflow automation, dashboards, database/API implementation, local stakeholder coordination and transition	Associate Degree in Computer & IT plus BSc Organizational Leadership; JavaScript, Node.js, React, PostgreSQL/MySQL and REST APIs; digital systems work since 2023
Tonieh Alpha Gwoah — Key Expert 3: Trainer, Data Support & Field Operations Coordinator	Training, user orientation, beneficiary-data support, field logistics, documentation, issue resolution and adoption	BBA plus computer coursework; CHF beneficiary assessment/training and programme-data oversight; WTO/UNOPS/EIF, World Bank/IDA and WFP-linked experience
Specialists mobilized as required	GIS, cybersecurity/privacy, DevOps/infrastructure, database/data migration, M&E/business analysis, QA and help desk	Named CVs and availability to be included in final submission where scored or required

6.1 Staffing Schedule — TECH-6

Role	M1	M2	M3	M4	M5	M6	M7
Project Manager / Technical Lead	F	F	F	F	F	F	F
Software Developer / Implementation Lead	F	F	F	F	F	F	P
Business/Data Lead	F	F	F	P	P	P	P
GIS Specialist	P	F	F	F	P	F	P
Security/DevOps Lead	P	F	P	F	F	F	P
QA/Test Lead	P	P	P	F	F	F	P
Trainer/Data Support/Field Operations	P	P	P	P	F	F	F

F = full-time input during the month; P = part-time/intermittent input. Exact staff-month totals and location breakdown will be finalized in the prescribed TECH-6 form after confirmation of named specialists and FAO's duration clarification.

7. Capacity Development and Knowledge Transfer

Capacity development will be based on tasks users must perform, not generic system demonstrations. TOTAG will conduct a training needs assessment and develop curricula for leadership, administrators, data stewards, county/district users, enumerators/supervisors, extension and programme users, GIS/analysts, help desk, infrastructure/security teams and development partners.

Audience	Learning focus	Method / evidence
National leadership/product owners	Governance, dashboards, policy use, acceptance and sustainability	Executive sessions; decision scenarios; action plan
Administrators/data stewards	Configuration, users/roles, reference data, quality, audit, retention and reporting	Hands-on lab; practical assessment; admin manual

Audience	Learning focus	Method / evidence
Enumerators/supervisors	Consent, identity, forms, GPS/parcels, offline work, sync, quality and safeguarding	Field simulation; observed competency; job aids
County/district/extension/programme	Verification, visits, eligibility, service workflows, reports and grievances	Role scenarios; exercises; coaching
Technical team	Architecture, APIs, deployment, database, monitoring, backup/restore, security and release management	Technical labs; runbook drills; recovery test
Help desk	Ticket intake, diagnosis, escalation, knowledge base and service metrics	Case simulations; escalation exercise

- Training-of-trainers with co-facilitation and observed delivery.
- Pre/post assessments and practical competency checks.
- User manuals, administrator guide, API guide, deployment/runbook set, quick-reference cards and short demonstration media.
- Post-training clinics, pilot coaching and hypercare support.
- Handover register covering source code, repositories, configurations, credentials process, licenses, infrastructure, test evidence and documentation.

8. Sustainability, Support and Ownership

The design will minimize recurrent complexity and ensure government teams can operate and extend the platform. Sustainability decisions—including hosting, software dependencies, map services, messaging, device management, certificates, backup and support—will be documented with ownership, renewal requirements and operational effort.

- Government ownership of approved data, configurations, documentation and project source code subject to final contract terms.
- Open, documented APIs and portable data formats.
- Configuration-driven forms, workflows, reference data, roles and programme rules.
- Environment, deployment and recovery automation with tested runbooks.
- Tiered support model: user self-service/job aids, help desk, application support and specialist escalation.
- Operational dashboards for availability, errors, sync, API performance, backup, security and service levels.
- Warranty/support period and response targets to be aligned to the final contract and FAO clarification.

9. Bidder Qualifications and Partnerships — TECH-1

9.1 Corporate Identity

Item	Position / evidence
Legal entity	TOTAG GROUP OF COMPANIES LTD — legal registration documents to be attached in the prescribed eligibility section
UNGM registration	UNGM ID 1185811, evidenced by supplied portal screenshot
Local presence	Paynesville, Montserrado County, Liberia
HP relationship	HP Partner Portal organization, Partner ID 991002750397; supplied screenshots show Michael Gwoah as Partner Portal Administrator
Bid contact	info@totaggroup.com

ACCURATE HP REPRESENTATION The supplied evidence supports HP Partner Portal registration and administrator access. It is not represented here as an HP manufacturer authorization letter, product quotation, server commitment, exclusivity, or guarantee for this tender. Any such claim must be supported by a separate current document issued or verifiable through HP.

9.2 Relevant Corporate Experience

The RFP requires documentary evidence of comparable corporate assignments. No signed contracts, completion certificates, client references or acceptance evidence were supplied with the materials used for this draft. The final TECH-1 form must contain only verifiable assignments and must attach the evidence required by the RFP.

Assignment / client	Scope and period	Contract value	Evidence
[Insert verified comparable assignment 1]	[Insert system development, training, deployment and support details]	[Financial data may appear only where the FAO form requires it; confirm separation rule]	[Contract / PO / completion / reference]
[Insert verified comparable assignment 2]	[Insert integration/interoperability and government/UN details]	[As required by prescribed form]	[Contract / PO / completion / reference]

9.3 Proposed Technology and Supply Ecosystem

TOTAG will use fit-for-purpose, supportable technologies selected during design and approved by FAO/MoA. The architecture is intentionally modular: enterprise server and storage; virtualized/containerized application environments; relational/spatial database; responsive web and offline mobile clients; API management; monitoring; backup and recovery. TOTAG has formally requested a tender-specific manufacturer/authorized-reseller letter from HP. Until issued, the available evidence supports HP Partner Portal registration only; server sizing, availability, warranty, licensing and authorized supply will be confirmed through HP or an authorized distribution channel.

10. Mandatory Compliance and Submission Readiness

The following matrix distinguishes a strong technical narrative from a compliant submission. Items marked "Action required" must be closed before upload. They are not optional editorial improvements.

Requirement	Current evidence	Status / action
Legal registration and eligibility declarations	Not included in reviewed attachments	ACTION REQUIRED — attach current registration, tax/eligibility and signed FAO declarations
Comparable corporate contracts / five years' experience	No contract evidence supplied	ACTION REQUIRED — complete TECH-1 with at least the RFP minimum and attach proof
Key Expert 1: Project Manager	V. Karmo holds a BSc in Computer Information Systems and has led public-sector software modernization for the Indiana General Assembly; the RFP says UN/INGO experience is preferred but "not limited to" those organizations	STRONG ALIGNMENT — attach signed TECH-5, degree and reference evidence emphasizing software project-management tasks
Key Expert 2: Software Developer	M. Gwoah combines Computer & IT education, organizational leadership and custom software/digital workflow experience since 2023	PARTIAL DOCUMENTARY GAP — attach portfolio/repository/client evidence; add explicit Android/iOS, hosting and HP-server specialist evidence where available

Requirement	Current evidence	Status / action
Key Expert 3: Trainer/Data Support	T. Gwoah evidences beneficiary training, programme data, customer support and donor/UN-linked operations	PARTIAL DOCUMENTARY GAP — attach training references and computer coursework; strengthen Android remote/on-site support evidence or add specialist coverage
Availability and signed FAO-format CVs	CVs supplied, not in prescribed TECH-5 format and not signed for this bid	ACTION REQUIRED — convert, obtain declarations/signatures and attach certificates
HP relationship	Partner Portal Admin and Partner ID screenshots; tender-specific authorization requested	PENDING — attach the issued HP authorization, recommended configuration, warranty and supply evidence when received
UNGM registration	Portal screenshot showing ID 1185811	SUPPORTED — include clean evidence and verify account profile is current
Duration	RFP contains 5-, 6- and 7-month references	CLARIFICATION NEEDED — proposal uses seven-month TOR maximum
Technical/financial separation	This document contains no price	COMPLIANT — keep all pricing and commercial totals in the Financial Proposal file

10.1 RFP Anomalies to Resolve Without Changing the Offer

- Completion period appears as five, six and seven months in different sections.
- Experience periods and contract-count language are not fully consistent across qualification sections and forms.
- TECH-7 contains content that appears unrelated to the DFR assignment; TOTAG should follow the DFR TOR and seek confirmation of the intended schedule format.
- Server quantities, final sizing, supply responsibility, hosting model, warranty/support, pilot scope, data volumes, migration sources and final acceptance criteria require confirmation.
- The final contract, intellectual-property terms, data-processing obligations, support period and payment-linked acceptance should be reviewed when issued.

11. Comments and Suggestions on the TOR — TECH-2

Topic	Constructive recommendation
Duration	Confirm a single contractual duration and align TECH-6/TECH-7 and payment milestones.
Hosting/infrastructure	Confirm target data centre/hybrid-cloud policy, existing government capacity, server quantity, sizing basis, warranty and who procures connectivity, backup and disaster-recovery resources.
Data and migration	Provide source-system inventory, estimated farmer/household/parcel volumes, data-quality samples, migration authority and retention rules.
Interoperability	Identify priority systems, owners, available specifications/sandboxes, data-sharing agreements and acceptance responsibility.
Pilot/rollout	Confirm pilot counties/districts, target users/devices, registration targets, rollout waves and field-logistics responsibility.
Security/privacy	Confirm applicable national policies, data-controller/processor roles, hosting restrictions, security assessment requirements and breach notification process.
Languages/accessibility	Confirm required languages, USSD/SMS scope, shortcode/MNO arrangements and accessibility standards.
Acceptance/support	Define deliverable review periods, defect severity thresholds, warranty/support period, service levels and final handover/IP terms.

12. Assumptions and Dependencies

- FAO and Government counterparts will nominate empowered product, data, integration, infrastructure, security and acceptance focal points.
- Authorized access to relevant systems, documents, sample data, facilities and stakeholders will be provided according to the approved inception plan.
- Data-sharing and legal authorizations will precede live exchange of personal or sensitive data.
- Pilot and rollout logistics, devices, connectivity, travel, venues and server procurement responsibilities will be agreed in the contract/inception plan.
- Third-party fees—including map, messaging, identity, MNO, payment, certificate, cloud or proprietary software services—will be governed by approved architecture and the separate Financial Proposal.
- No biometric processing will be implemented without explicit scope, legal basis, impact assessment, security design and approval.
- Changes after design approval will be controlled through documented impact assessment and authorization.

13. Conclusion

TOTAG's proposal is designed to help Liberia establish a living national digital service: trusted enough for beneficiary decisions, practical enough for county and district field use, interoperable enough to connect programmes, and maintainable enough for long-term government ownership. Our stage-gated approach reduces risk, makes progress visible, and places data governance, security, field realities and capacity transfer at the centre of delivery.

Subject to closure of the mandatory evidence actions identified in Section 10, TOTAG is prepared to mobilize promptly after contract signature and work jointly with FAO, the Ministry of Agriculture and partner institutions to deliver the complete scope.

Annex A — CV Evidence Summaries

A1. M. Tarwoyouberkowu Gwoah

- BSc, Organizational Leadership & Supervision, Purdue School of Science & Technology (2012).
- Associate Degree, Computer & Information Technology, Purdue School of Science & Technology (2011).
- Founder and CEO, TOTAG GROUP OF COMPANIES LTD; executive coordination, governance, reporting and stakeholder engagement.
- Software Developer & Technology Consultant (2023–present) focused on workflows, dashboards and administrative systems.
- Proposed as Project Director/Institutional Lead; mandatory Project Manager compliance requires additional evidence or a separate compliant nominee.

A2. Varnie N'Jola Karmo

- BSc, Computer Information Systems, Purdue University (1992).
- More than 25 years of enterprise software engineering and data architecture experience.
- Senior Developer, Indiana General Assembly Office of Technology Services (since 2007).
- Experience includes React, PostgreSQL, AWS, Oracle, Neo4j, Jasper, legacy modernization, integration and GIS polygon matching.
- Proposed as Lead Software Architect/Developer; updated evidence is required for the RFP's explicit mobile-platform, mobile-training and HP-server-procurement criteria.

A3. Tonieh Alpha Gwoah

- BBA, Management & Economics, University of Liberia (2007), with computer-science coursework.
- General Manager, TOTAG GROUP OF COMPANIES LTD; operations, staff supervision, procurement and reporting.
- Business Development Officer, CHF International Liberia YES Project; beneficiary assessments, targeted training, data and grants/procurement compliance.
- Additional programme support experience with WTO/UNOPS/EIF, World Bank/IDA, WFP and LBDI.
- Proposed as Operations, Stakeholder Engagement & Training Coordinator; a separate compliant Trainer/Data Support key expert should be nominated unless FAO confirms equivalence.

Full signed CVs in FAO TECH-5 format, degrees/certificates, references and availability declarations should follow this annex in the final Technical Proposal file.

Annex B — Documentary Evidence Register

Evidence	Reviewed	Final submission action
HP Partner Portal Administrator email screenshot	Yes	Include clean copy; do not overstate
HP portal Partner ID 991002750397 screenshot	Yes	Include clean copy; obtain current product/supply authorization if relied upon
UNGM portal screenshot, ID 1185811	Yes	Include clean copy; verify profile status
Three personnel CVs	Yes	Convert to TECH-5, strengthen evidence, sign and attach certificates
Legal registration / tax / eligibility	No	Obtain and attach
Comparable corporate contracts and completion evidence	No	Obtain and attach
Expert references and availability	Partial	Obtain signed declarations and current contacts
Consortium/subcontractor letters, if any	No	Use only if applicable and comply with RFP designation/acceptance requirements

Annex C — Final Technical File Assembly Checklist

- Signed technical proposal letter and prescribed Proposal Summary/eligibility declarations.
- TECH-1 corporate experience, completed with verifiable contracts and required supporting evidence.
- TECH-2 comments and suggestions.
- TECH-3 approach, methodology and work plan.
- TECH-4 team composition and assignments.
- TECH-5 signed CVs with degree/certificate and experience evidence.
- TECH-6 staffing schedule with staff-months, home/field input and signatures where required.
- TECH-7 work schedule aligned to FAO's confirmed duration.
- Legal registration, UNGM evidence and any required certifications/declarations.
- HP evidence described accurately; manufacturer/reseller/quotation evidence if material to the offer.
- No pricing, rates, totals or commercial conditions in the Technical Proposal file.
- Separate Financial Proposal prepared strictly in the prescribed forms and currency.
- PDF is searchable, page-numbered, legible, correctly named and uploaded before 4 August 2026 at 12:00 Monrovia time.

